

CONSOLIDATED INTELLIGENCE PLATFORM

Why This Approach Works

A plain-language look at what we built, how it helps investigators, and the evidence behind it

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What We Built, In One Paragraph

Police investigators already collect FIRs, call records, bank transaction records, surveillance notes, and intelligence-agency reports on every case. The problem was never a lack of data - it was that the data sits in separate systems, in free text and spreadsheets, and connecting it by hand is slow enough that organised networks can hide inside the gaps. We built a system that reads all five of those record types, automatically works out who is really who, draws the relationship map between people, phones, accounts, vehicles and locations, and highlights the specific patterns organised crime uses to look fragmented on paper - so an investigator spends their time verifying leads, not assembling them from scratch.

Three things we want to be upfront about

- **It is a live, working system today** - not a slide deck or a concept. Every number in this report comes from running it, not from an estimate.
- **It never decides guilt.** It produces leads, each one traceable back to the exact records behind it, and every lead requires a human investigator's sign-off before it means anything.
- **We report our own weaknesses honestly**, including a genuinely low accuracy number on our hardest feature (Section 6) - because a system that only shows its best numbers isn't one an investigator should trust with real cases.

Why This Is Genuinely Hard Today

This system responds to SIH26189, a problem statement about uncovering hidden criminal networks from fragmented data. In plain terms, the problem statement describes something every investigator already knows from experience:

- A single case might touch an FIR, a dozen call records, a handful of bank transfers, and a surveillance note - each filed in a different system, by a different person, in a different format.
- Organised networks deliberately use intermediaries, mule accounts, and burner-SIM swaps specifically because these tactics make the paper trail look like unrelated, disconnected activity.
- Manually cross-referencing hundreds of records to notice that two "unrelated" phone numbers are actually the same person, or that one account is quietly bridging two separate fraud rings, does not scale - and the people who benefit most from that difficulty are the ones running the network.

The problem statement asks for a system that ingests this fragmented data, extracts the people/places/organisations inside it, builds a relationship map, identifies who is actually influential (not just who appears most often), flags suspicious patterns, and hands all of that to a human investigator as usable intelligence. That is exactly what we built - point for point, covered in Section 3.

Every Requirement, Answered

What the problem statement asks for	What we built
Collect and process data from multiple sources	FIRs, surveillance reports, intelligence-agency reports, call records, and financial transactions - five sources, all live-ingested
Extract people, locations, vehicles, phone numbers, organisations	An extraction pipeline combining AI text-reading (NLP) with rule-based backups specifically added because the AI model alone missed names in short, terse sentences during testing
Build relationship maps	An interactive graph showing every person/phone/account/vehicle/location and how they connect, with the type of evidence behind every connection visible on click
Identify key influential individuals	Network-analysis scoring (the same maths used to rank influence in social networks) tuned specifically so a busy customer-care number or an investigating officer's own phone can never be mistaken for a kingpin
Detect suspicious patterns	Five purpose-built detectors: mule-account layering, burner-SIM rotation, a call immediately before a large transfer, financial "bridge" accounts linking two unrelated networks, and trafficking recruiter/transporter chains
Assist investigators with visual and analytical insight	A full investigator dashboard: case overview, interactive graph, a lead list with plain-language explanations, an identity-review queue, and a self-grading dashboard

One line we hold to throughout: every score in this system attaches to a **lead** or a **connection** between two entities - never to a person as a standalone risk number. That distinction is not a technicality; it is the difference between a tool that assists an investigation and one that could be misused to pre-judge a person.

Built-In Safeguards, Not Afterthoughts

A system like this is only as good as an investigator's ability to trust what it tells them. We treated that as a design requirement from day one, not a compliance checkbox added later.

1. It never accuses - it evidences

Every single finding the system produces links back to the specific FIR, call record, or transaction it came from. An investigator can always click through and read the actual source text or record - nothing is a black-box score with no explanation attached.

2. A human always has the final word

Every lead sits in a queue with three buttons: Useful, Already Known, Wrong Person. The system does not remove itself from that queue, does not escalate itself, and does not take any automated action of any kind. It surfaces; a person decides.

3. Every action is permanently, verifiably logged

Every query and every decision is chained together with cryptographic hashes - the same core idea blockchains use for tamper-evidence - so that altering any past entry would break every hash that comes after it. This is checked twice, independently: once by the server, and once by the investigator's own browser, so no single component has to be blindly trusted.

4. When the system is unsure, it says so - it doesn't guess

When two records mention a similar name but there isn't enough shared evidence (a phone number, an account) to confirm they're the same person, the system does not silently merge them. It puts the ambiguous mentions in a review queue for a person to resolve - because a wrong merge in a criminal case is far worse than an unresolved question.

We Grade Our Own Work - Including the Weak Parts

The system includes a live self-evaluation dashboard that checks its own accuracy every time it runs, against a checklist of known scenarios. We are showing the actual current numbers below, not a best-case run.

What we checked	Result
Correctly keeps two different people with the same name separate	YES
Correctly merges one person who appears under two name spellings	YES
Never lets an investigating officer's own phone top a ranking	YES
Never lets a customer-care number top a ranking	YES
Detects a burner-SIM rotation	YES
Detects a mule-account layering chain	YES
Detects a call immediately preceding a large transfer	YES
Finds the recruiter in a trafficking chain	YES
Finds the transporter/intermediary in a trafficking chain	YES

The one number we are not proud of - and why we're showing it anyway

One experimental feature ("missing-link recovery" - suggesting a relationship that might already exist but wasn't directly recorded) currently recovers only a small fraction of genuinely missing connections when tested honestly: we hide 20% of real connections, ask the system to guess them back blind, and measure how many it actually finds.

Metric	Value
Connections hidden for the test	48
Recovered in the top 10 guesses	2.1%
Recovered in the top 20 guesses	4.2%
Recovered in the top 50 guesses	4.2%

We could have hidden this number, or tuned the test until it looked better. We didn't, for two reasons: first, because a large share of the hidden connections in our test data are single, isolated links with no surrounding pattern for any method to reconstruct - which is an honest limitation of the underlying approach, not a bug to paper over; and second, because a system that only ever shows its best numbers is exactly the kind of system an investigator should be skeptical of. We would rather earn trust with one weak, clearly-labelled number than lose it by hiding one.

Validated Against Real, Cited Cases

All data inside the demo is synthetic - fabricated names and numbers generated for this presentation, with no real person's information anywhere in the system. But the patterns it is built to catch are not hypothetical. We checked our detectors against two real, recently reported law-enforcement operations.

A financial network - CBI Operation Chakra-VI (September 2026)

A coordinated raid across 89 locations in 20 states arrested three people playing exactly the three roles our mule-layering detector is built to separate: one account that received funds, one that "routed" (layered) them through a business account, and one that withdrew and transferred the remainder. A separate CBI release from the same operation used the word "kingpin" for the network's head - precisely the framing our influence-scoring is built around.

"the CBI arrested three persons allegedly involved in receiving, layering and dispersing fraud proceeds" - Coin Edition, 5 September 2026

A trafficking network - Delhi Police / J&K Police (August 2025)

Delhi Police's Anti-Human Trafficking Unit, working with J&K Police, dismantled a network trafficking an estimated 500 people over two years. The roles investigators identified - a recruiter running an agency, a named transporter who ferried victims from a specific railway station on instructions from agents, and a receiving end in another state - map directly onto the Recruiter → Transporter → Receiver chain our Women-Safety detector is built to find.

"Suraj allegedly admitted he ferried trafficked persons from railway stations in Delhi on the instructions of agents" - Hindustan Times, 28 August 2025

Sources: Coin Edition, "Indian CBI Raids 89 Locations Across 20 States Under Operation Chakra-VI" (coinedition.com, 5 Sep 2026); News On Air / Akashvani, "CBI Arrests Kingpin of Transnational Cyber Fraud Syndicate" (newsonair.gov.in, 25 Aug 2025); ANI, "Delhi Police busts human trafficking racket, 4 arrested" (aninews.in, 28 Aug 2025); Hindustan Times, "Delhi Police busts interstate trafficking network in Srinagar" (28 Aug 2025).

Protecting Women and Children Is Not a Side Feature

This problem statement is filed under NCRB's Women Safety Division, and we treated that as a signal about what matters most, not a footnote. The trafficking-network view is built to the same standard as the general financial-crime view - in fact, it is **the same underlying machinery**, re-applied to the person/phone network instead of the account network. The exact same code that finds a financial intermediary finds a trafficking transporter.

We also found, and engineered around, a real limitation rather than hiding it: the standard community-detection method used for financial networks does not reliably separate a small, tightly-connected trafficking chain (recruiter, transporter, receiver) into distinct groups - it tends to see the whole chain as one blob. Rather than claim a method works when we had evidence it doesn't for small chains, we built and shipped a second, independent method specifically for this case, and we run both, clearly labelling which one found each result.

We also added a signal not originally in most systems like this: when the **same real-world location** is named across several independent reports tied to different people, that recurrence itself is meaningful - it is often how a trafficking corridor gets noticed in the first place, and it is exactly what happened in the real Delhi/Srinagar case above (victims recruited repeatedly from the same railway station).

It's Live - You Can Open It Right Now

Everything described in this report is running on the public internet today, not staged for a single demo. We are giving you the same access we use ourselves.

What	Where
Live application	the deployed frontend URL provided separately
Backend service	the deployed backend URL provided separately
Source code	the GitHub repository provided separately

Account	Password	Access
investigator1	investigator1pass	The financial-fraud case
investigator2	investigator2pass	The trafficking case
admin1	admin1pass	Both cases, plus the audit-tamper demonstration

We also built an automated test suite that checks every one of the planted scenarios listed in Section 5 against the live system every time it changes, and an in-app guided tour that walks a first-time visitor through every screen without needing anyone to narrate it. Neither of those exists because a hackathon expects them - they exist because a tool investigators are meant to rely on has to keep working correctly as it grows, and has to be usable without a training session.

Where We'd Take This, Honestly

We would rather tell you what isn't finished than let a polished demo imply it is.

- **Real data would need a small adapter, not a rebuild.** The analysis engine already treats every record type generically; connecting a real department's FIR/CDR/transaction exports means writing one import script per source, not rewriting the system.
- **The database would need to move to a production-grade system** (we use a lightweight, easily-inspected database for this stage deliberately, so the logic stays transparent and auditable while it's being evaluated).
- **Missing-link recovery needs more data to prove itself properly** - the honest low number in Section 6 reflects a sparse test set as much as the method itself, and a larger real dataset would be a fairer test.
- **Any real deployment needs a conversation with legal/IT stakeholders first** about data protection, retention, and access policy - the role-based access and audit trail here are a foundation for that conversation, not a substitute for it.

We built this to be judged on what it actually does, not on what a slide claims it does. Every figure in this report came from running the live system on the date above - we'd welcome you trying it yourself.